

Scope: Post installation steps and Configuration ARA (Access Risk Analysis)

Target audience: SAP GRC consultants

Pre-requisite: Need to have good understanding of how sap security And SOD (Segregation of duties).

Steps 1:

We need to create a connector nothing but the RFC connection to the backend systems Through SM59 t-code.

The screenshot shows the SAP SM59 'RFC Destination' configuration window for 'grcclnt001'. The window has a menu bar (Connection, Edit, Goto, Extras, Utilities(M), System, Help) and a toolbar. The main area is divided into several sections:

- Remote Logon** tab is selected. It shows:
 - RFC Destination: grcclnt001
 - Connection Type: 3 ABAP Connection
 - Description: target sys
- Administration** tab is selected. It shows:
 - Target System Settings**:
 - Load Balancing Status: Load Balancing is set to No (radio button).
 - Target Host: india.server.com
 - System Number: (empty)
 - Save to Database as: Save as is set to IP Address (radio button), with the value 200.200.200.200.
 - Gateway Options**: Gateway Host and Gateway service fields are empty, with a Delete button next to them.

Once done do the test connection.

RFC - Connection Test

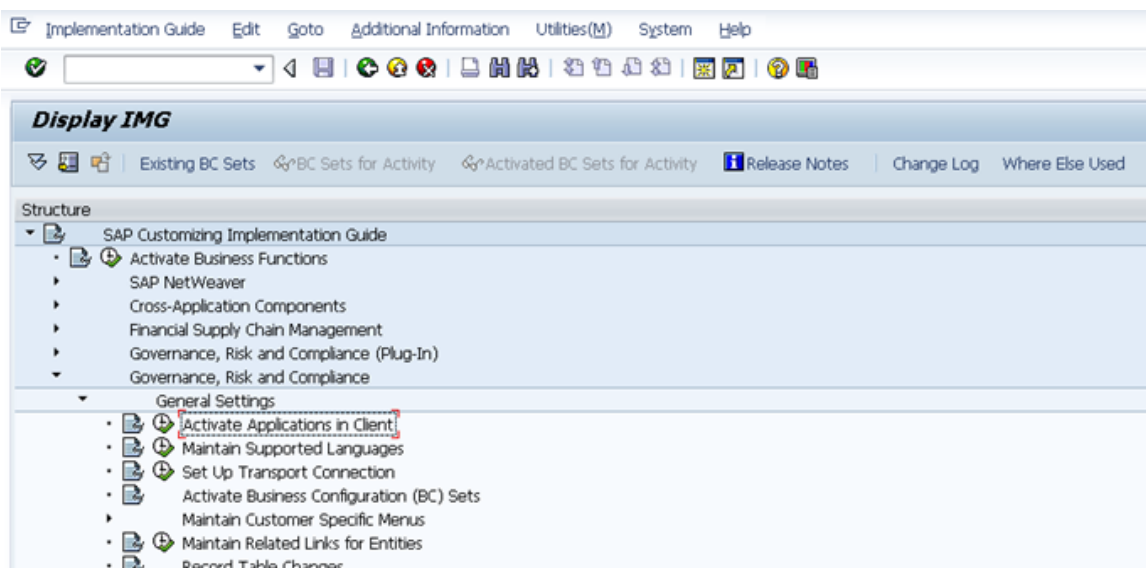
Connection Test grcclnt001
Connection Type SAP Connection

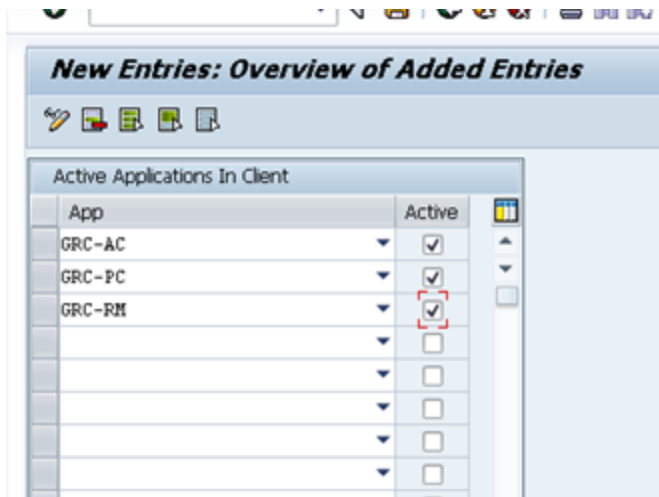
Action	Result
Logon	6 msec
Transfer of 0 KB	0 msec
Transfer of 10 KB	1 msec
Transfer of 20 KB	1 msec
Transfer of 30 KB	1 msec

Step: 2: Activate the application in Client.

This step is need to activate the application in client like AC, PC ,RM as they are all resides in one GRC box.

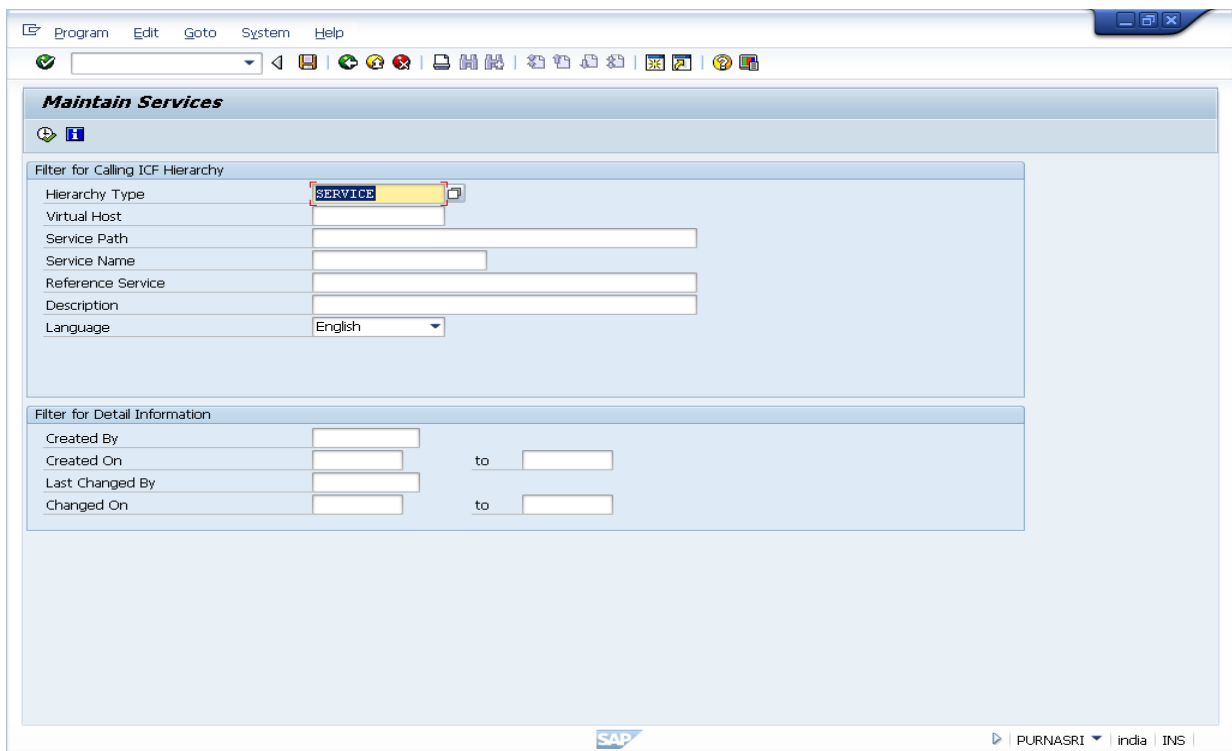
Go to: SPRO→GRC→General settings→ Activate the application in Client

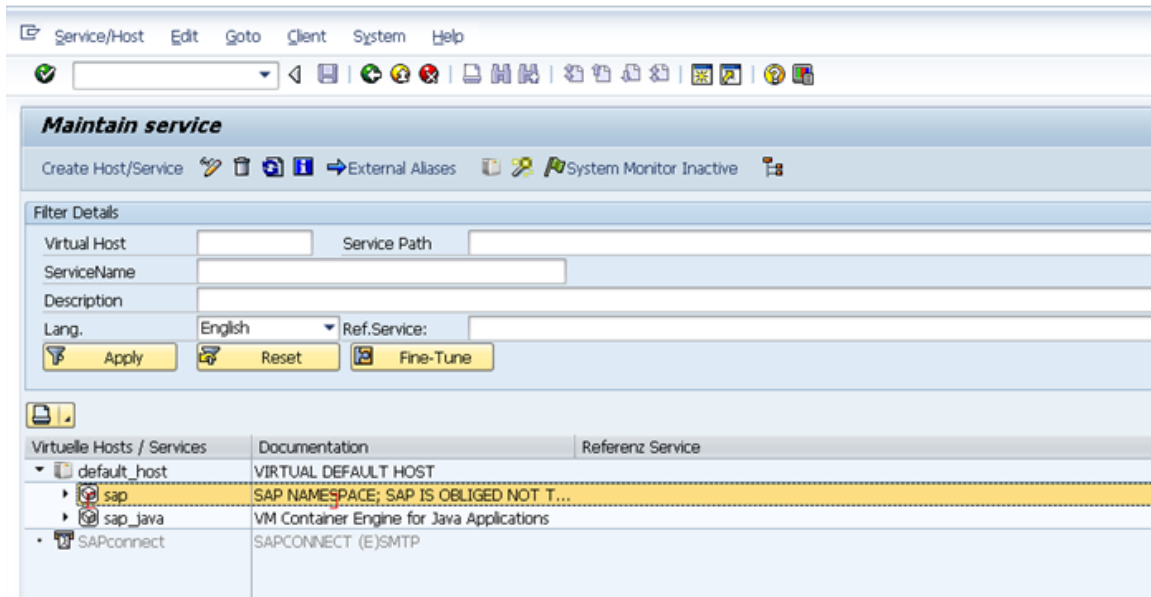




Step3: Activating the services

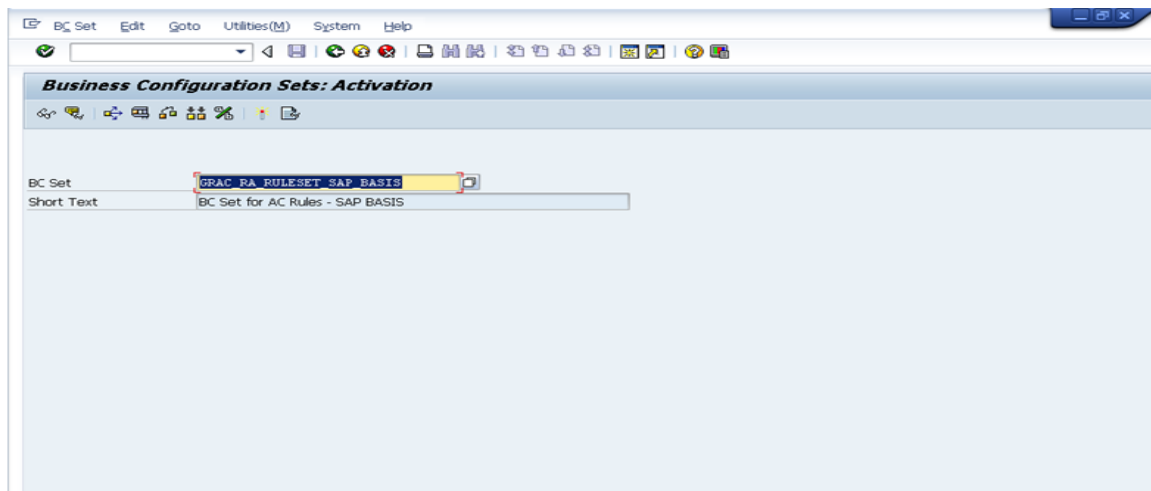
SICF: activate HTTP services: Which is used to access Portal and NWBC and Web dynpro Screens.





Step4: Activating the BC Sets (Business configurations)

Go to: SCPR20---Select the required BC sets which you want to activate.



Business Configuration Set (1)

Search by BC Set Attributes

BC Set

Short text

Changed by

Changed on

Software Component

Release

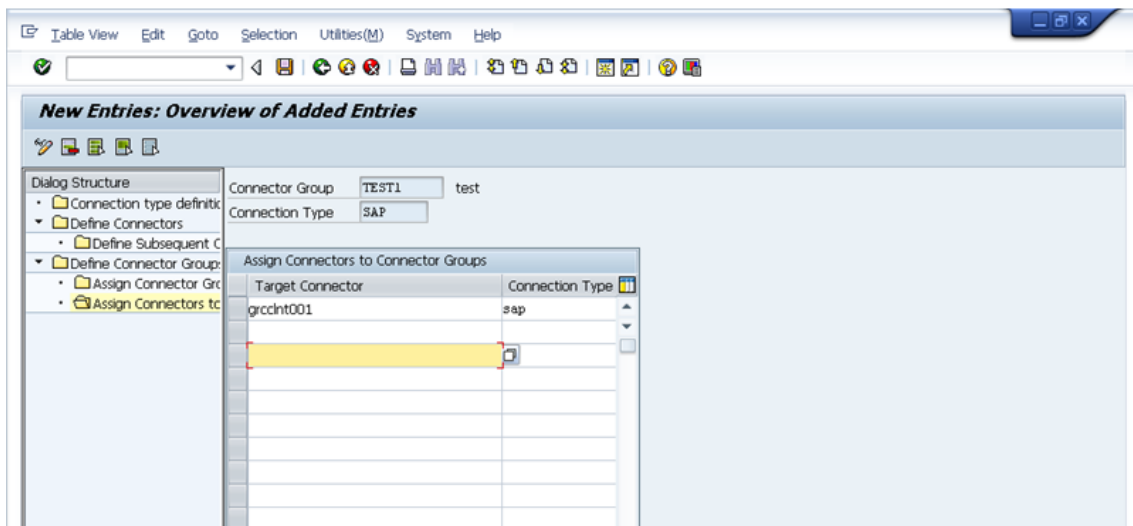
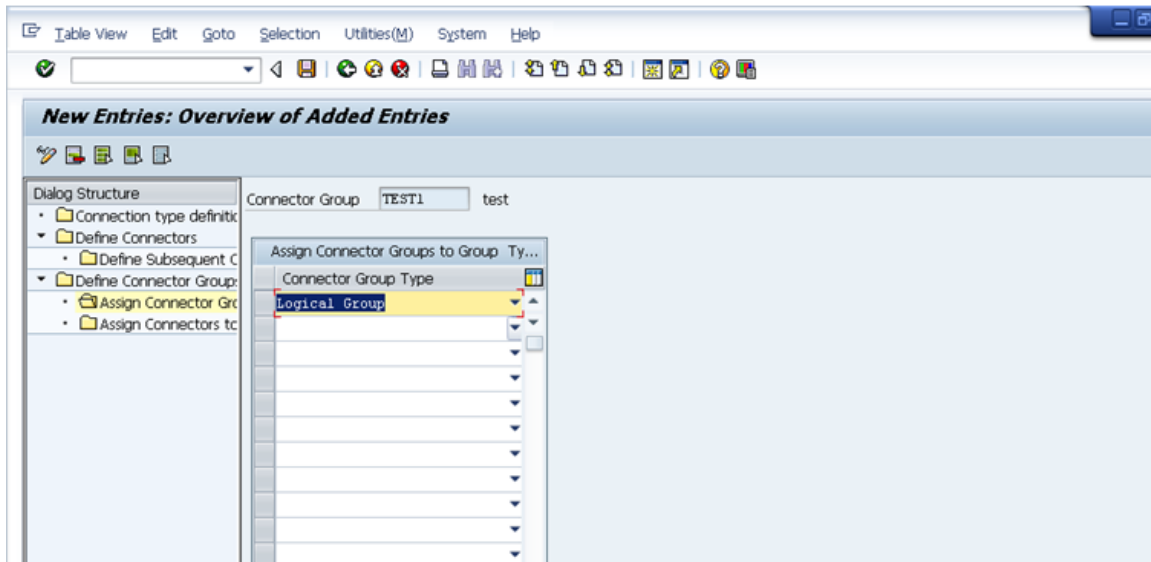
Maximum No. of Hits

Business Configuration Set (1) 22 Entries found

Search by BC Set Attributes

BC Set ID	BC Set description	Changed by	Changed	Software Compo...	Relea...	Relea...
GRAC_ACCESS_REQUEST_APPL_MAPPING	BC Set for Access Request - Application function mapping	ATWALM	20.08.2010	GRCFND_A	V1000	*
GRAC_ACCESS_REQUEST_EUP	BC Set for Access Request - End User Personalization	AGARWALSAN	28.04.2011	GRCFND_A	V1000	*
GRAC_ACCESS_REQUEST_PRIORITY	BC Set for Access Request - Priority	ATWALM	19.08.2010	GRCFND_A	V1000	*
GRAC_ACCESS_REQUEST_REQ_TYPE	BC Set for Access Request - Request Type	MISHRASE	12.01.2012	GRCFND_A	V1000	*
GRAC_RA_RULESET_COMMON	Rule Set for Common rules	ATWALM	30.06.2011	GRCFND_A	V1000	*
GRAC_RA_RULESET_JDE	BC Set for AC Rules for JDE	ATWALM	27.10.2010	GRCFND_A	V1000	*
GRAC_RA_RULESET_ORACLE	BC Set for AC Rules for ORACLE	ATWALM	27.10.2010	GRCFND_A	V1000	*
GRAC_RA_RULESET_PSOFT	BC Set for AC Rules for PeopleSoft	ATWALM	27.10.2010	GRCFND_A	V1000	*
GRAC_RA_RULESET_SAP_APO	BC Set for AC Rules - SAP APO	ATWALM	27.10.2010	GRCFND_A	V1000	*
GRAC_RA_RULESET_SAP_BASIS	BC Set for AC Rules - SAP BASIS	ATWALM	30.06.2011	GRCFND_A	V1000	*
GRAC_RA_RULESET_SAP_CRM	BC Set for AC Rules for SAP CRM	ATWALM	27.10.2010	GRCFND_A	V1000	*
GRAC_RA_RULESET_SAP_ECCS	BC Set for AC Rules for SAP ECCS	ATWALM	27.10.2010	GRCFND_A	V1000	*
GRAC_RA_RULESET_SAP_HR	BC Set for AC Rules for SAP HR	ATWALM	30.06.2011	GRCFND_A	V1000	*
GRAC_RA_RULESET_SAP_NHR	BC Set for AC Rules for SAP R3 less HR Basis	ATWALM	30.06.2011	GRCFND_A	V1000	*
GRAC_RA_RULESET_SAP_R3	BC Set for AC Rules for SAP R3	BATRAJ	10.11.2011	GRCFND_A	V1000	*
GRAC_RA_RULESET_SAP_SRM	BC Set for AC Rules for SAP SRM	ATWALM	01.07.2011	GRCFND_A	V1000	*
GRAC_ROLE_MGMT_LANDSCAPE	BC Set for Business Roles Landscape	ATWALM	30.09.2010	GRCFND_A	V1000	*
GRAC_ROLE_MGMT_METHODODOLOGY	BC Set for Role Management Methodology Process and Steps	ATWALM	18.08.2010	GRCFND_A	V1000	*
GRAC_ROLE_MGMT_PRE_REQ_TYPE	BC Set for Role Management Pre Requisite Types	ATWALM	18.08.2010	GRCFND_A	V1000	*
GRAC_ROLE_MGMT_ROLE_STATUS	BC Set for Role Management Role Status	ATWALM	18.08.2010	GRCFND_A	V1000	*
GRAC_ROLE_MGMT_SENSITIVITY	BC Set for Role Management Sensitivity Level	ATWALM	18.08.2010	GRCFND_A	V1000	*
GRAC_SPM_CRITICALITY_LEVEL	Criticality Level	ATWALM	17.08.2010	GRCFND_A	V1000	*



Step 2 : Assign connectors to integration scenarios Like AUTH, PROV,SUPMG,ROLMG

Display IMG

Existing BC Sets | BC Sets for Activity | Activated BC Sets for Activity | Release Notes | Change Log | Where Else Used

Structure

- SAP Customizing Implementation Guide
 - Activate Business Functions
 - SAP NetWeaver
 - Cross-Application Components
 - Financial Supply Chain Management
 - Governance, Risk and Compliance (Plug-In)
 - Governance, Risk and Compliance
 - General Settings
 - Shared Master Data Settings
 - Reporting
 - Common Component Settings
 - Integration Framework
 - Create Connectors
 - Maintain Connectors and Connection Types
 - Maintain Connection Settings**
 - Maintain Service Providers and Consumer Proxies in SOA Manager

- IMG activity documentation
- Internal Audit Management

Integration Scenario (1) 5 Entries found

Restrictions

Integ.Sc...	Integ.Scenario Text
AM	Automatic Monitoring
AUTH	Authorization Management
PROV	Provisioning
ROLMG	Role Management
SUPMG	Super User Privilege Management

Table View | Edit | Goto | Selection | Utilities(M) | System | Help

Change View "Subscenario definition": Overview

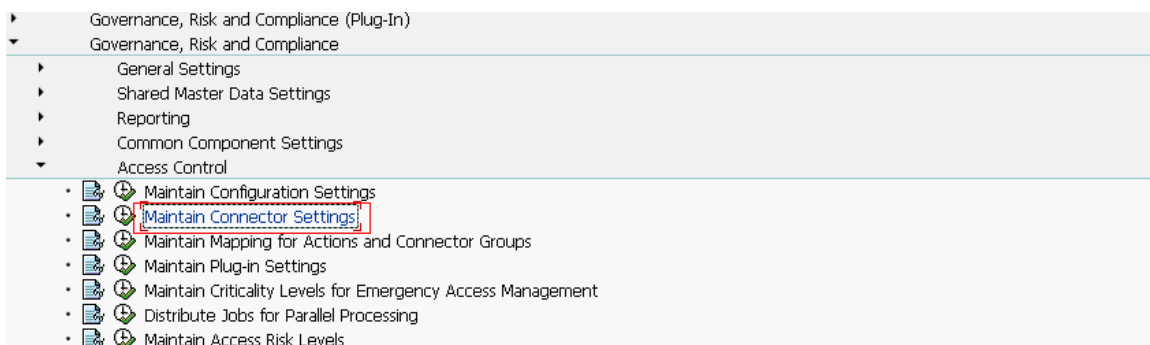
New Entries

Dialog Structure	Integration Scenario	AUTH Authorization Management
Subscenario definition	Subscenario definition	
Scenario-Connection type Link	Sub Scenario	Sub Scenario Text
Scenario-Connector L	AUTH	Authorization Management
Maintain file path:		

Risk Analysis	1023	01	Default report type for risk analysis
Risk Analysis	1024	1	Default risk level for risk analysis
Risk Analysis	1025	ELECTROL	Default rule set for risk analysis
Risk Analysis	1026	A	Default user type for risk analysis
Risk Analysis	1027	YES	Enable Offline Risk Analysis
Risk Analysis	1036	YES	Show All Objects in Risk Analysis
Risk Analysis - Spool	1051	200000	Max number of objects in a file or database record
Risk Analysis - Spool	1052	/usr/sap/D71/SYS/global	Spool File Location
Risk Analysis - Spool	1053	F	Spool Type

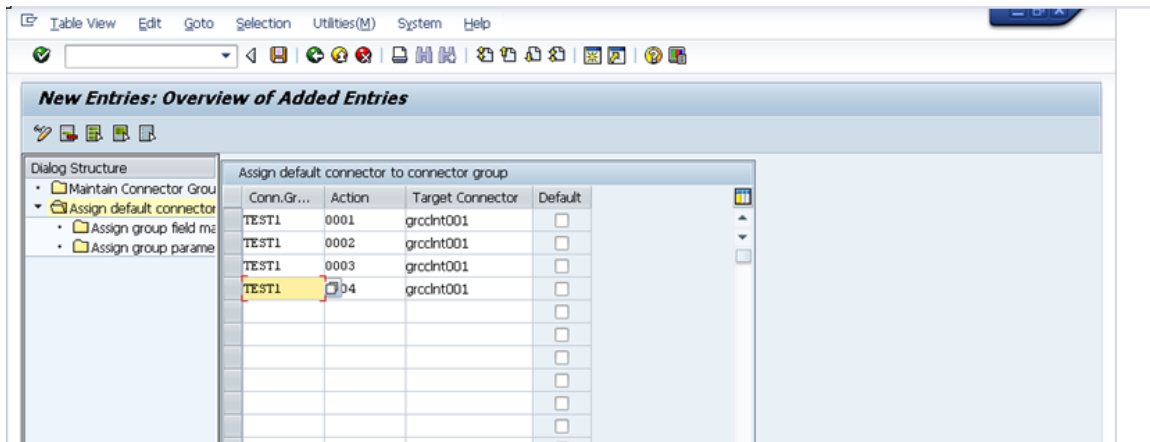
Step 2: Go to: SPRO→GRC→Access control→ Maintain connector setting

Here, we can maintain the connector settings for the systems that are connected to the access control application.



To add new connectors:

1. Choose the *New Entries* pushbutton.
2. From the respective dropdown lists, select the target connector, application type, and environment.
3. To set it as active, select the *Active* checkbox.



Once done with configuration of the connector settings.

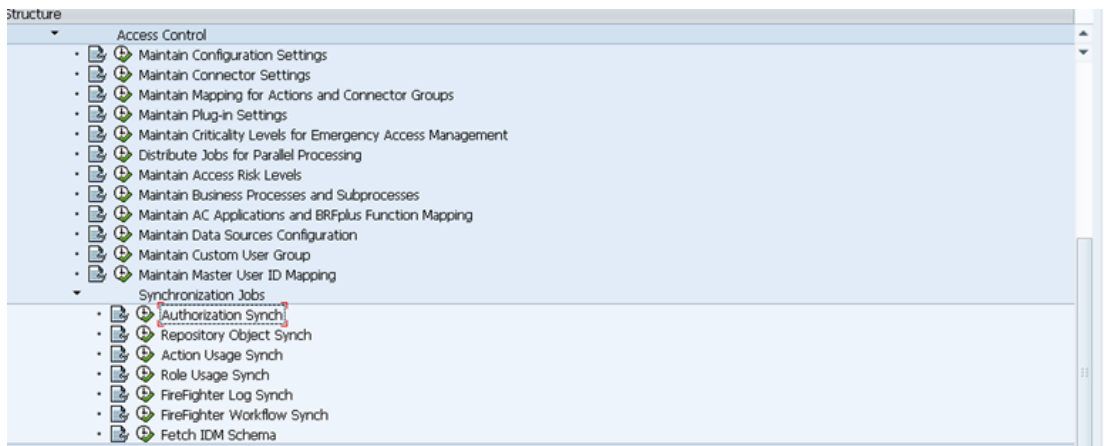
Step 4: Synchronization Jobs:

IMG -> Governance, Risk and Compliance -> Access Control -> Synchronization Jobs
-> Authorization Synch:

Here, we can synchronize the authorization master data from the back-end ERP systems, and store it in the GRC AC repository.

The synchronization updates the data for the following:

- Resource Sync - permissions, resources, and descriptions for authorization objects
- Action Sync - descriptions for actions, and permissions and resources for authorization objects
- Resource Class Sync - permissions and resources for authorization object classes and their relationships
- Resource Extension - organization level, activities, and descriptions for resource extensions
- Default (SU24) Values Sync - default authorization objects and field values for actions



Give the required connector to sync and then execute

Program for Authorization data Synchronization

Connector: [Yellow field] [Icon] [Arrow]

Language: EN to [Field] [Arrow]

For Legacy Systems

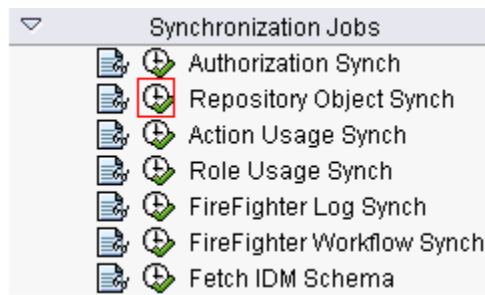
☐ Legacy System

To Connector: [Field]

Repository Object Synch:

IMG -> Governance, Risk and Compliance -> Access Control -> Synchronization Jobs
->

Repository Object Synch:



Here, we can synchronize the data for profiles, roles and users from the ERP back end and legacy systems and store it in the GRAC repository.

SOD Rules:

IMG -> Governance, Risk and Compliance -> Access Control -> Access Risk Analysis
-> SOD Rules

Here we can have the default SOD Rules due to activation of BC sets



Step 2:

IMG -> Governance, Risk and Compliance -> Access Control -> Access Risk Analysis
-> SOD Rules -> Upload SoD Rules:



Here, we can upload the GRAC SOD rules to the AC repository. You can choose to append to, or overwrite the existing data.
SAP provides standard file templates for uploading data related to business process, functions, risk and rule set.

The screenshot shows a software interface with two main sections. The top section, titled "System Selection", contains a dropdown menu labeled "System" with a checkmark icon. The bottom section, titled "File Selection", contains a list of items with checkboxes: "Business Process" (highlighted in yellow), "Function", "Function Business Process", "Function Actions", "Function Permissions", "Rule Set", "Risk", "Risk Description", and "Risk Rule Set Relationship". All checkboxes are checked.

Browse the files where you want to save and execute.

Step 3 : Generate SOD rules

IMG -> Governance, Risk and Compliance -> Access Control -> Access Risk Analysis
-> SOD Rules -> Generate SoD Rules

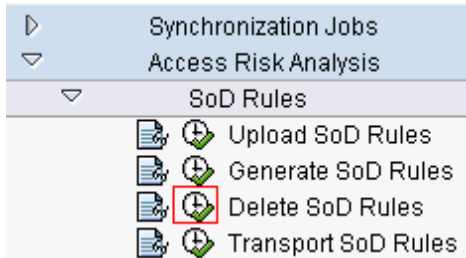


Here, we can generate the GRAC SOD rules for Risks.

The screenshot shows a software interface titled "Program for Generate Access Rules". Below the title bar, there is a section with a "Risk Id" label, a text input field containing "A001", and a "to" label followed by a dropdown menu showing "SDMD" and a checkmark icon.

Step 4 : Delete SOD Rules

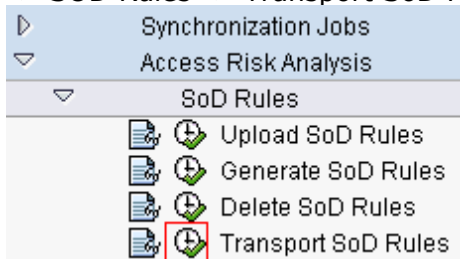
IMG -> Governance, Risk and Compliance -> Access Control -> Access Risk Analysis
-> SOD Rules -> Delete SoD Rules:



Here, we can delete the existing GRAC SOD rules.

Step 5: Transport SOD Rules:

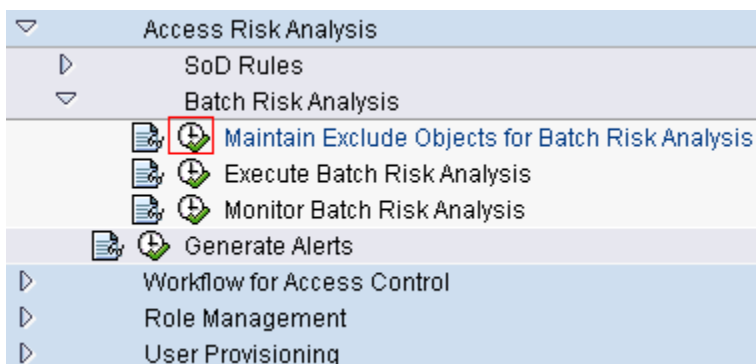
IMG -> Governance, Risk and Compliance -> Access Control -> Access Risk Analysis
-> SOD Rules -> Transport SoD Rules:



Here, we can create a transport for the GRAC SOD rules.

Maintain Exclude Objects for Batch Risk Analysis

IMG -> Governance, Risk and Compliance -> Access Control -> Access Risk Analysis
-> Batch Risk Analysis -> Maintain Exclude Objects for Batch Risk Analysis:



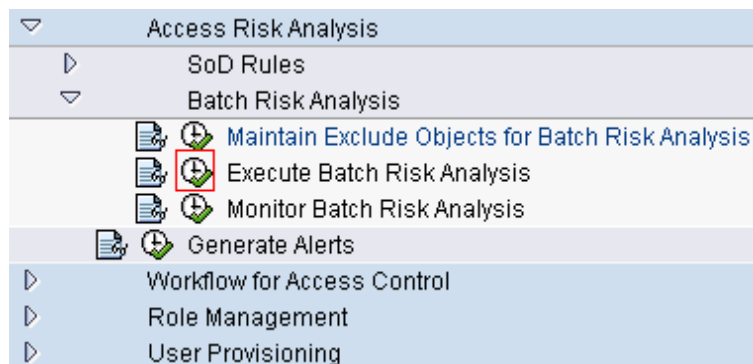
Here, we can maintain the list of objects you want to exclude from the batch risk analysis. By default, the application performs batch risk analysis on all objects in the specified system. You can choose to exclude specific objects that you know are safe or do not need to be analyzed.

Like SAP_ ALL, SAP_NEW profiles

Maintain excluded objects					
Obj ...	From Object	To Object	System	Status	Comments
Prof_ SAP_NEW			GRCLNT_	☒	Exclude sap_new profile
☒	☒		☒	☐	
☒	☒		☒	☐	
☒	☒		☒	☐	

Execute Batch Risk Analysis

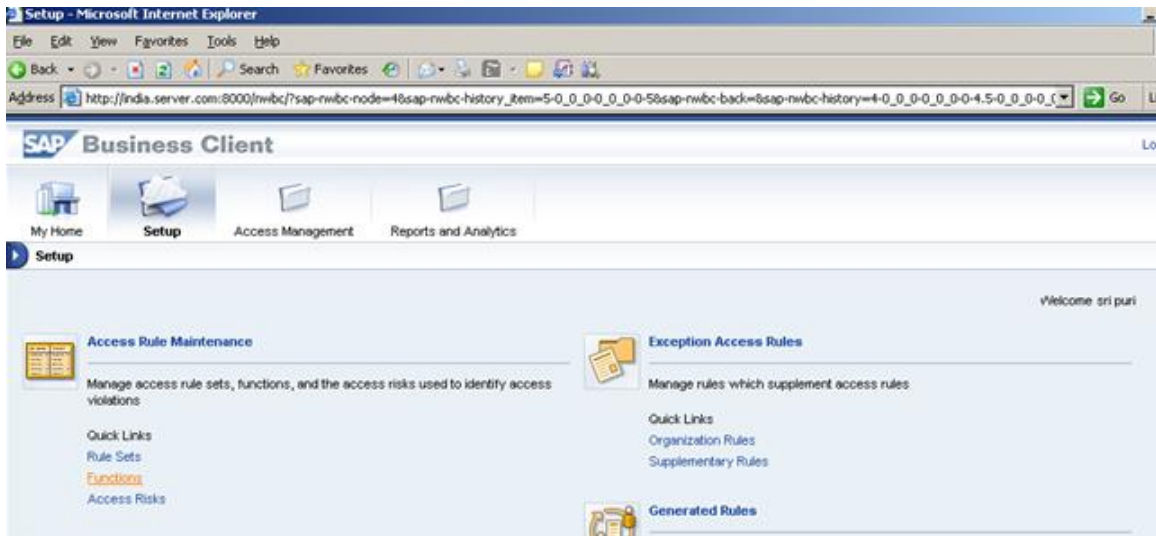
IMG -> Governance, Risk and Compliance -> Access Control -> Access Risk Analysis
-> Batch Risk Analysis -> Execute Batch Risk Analysis:



Here, we can schedule the Batch Risk Analysis. Batch Risk Analysis is required for Offline Risk Analysis and Management reports.

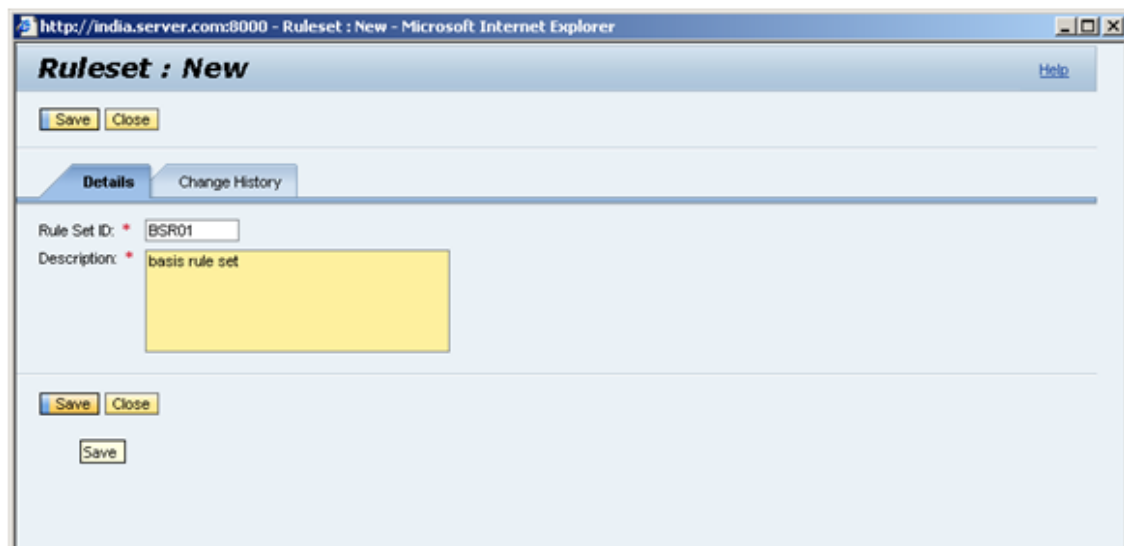
Creating new Ruleset, Functions and risks in NWBC

Logon to NWBC -> Choose setup work center -> *Access Rule Maintenance*



Step 1: Creating new Ruleset.

Choose setup work center->*Access Rule Maintenances* → *Rulesets*.

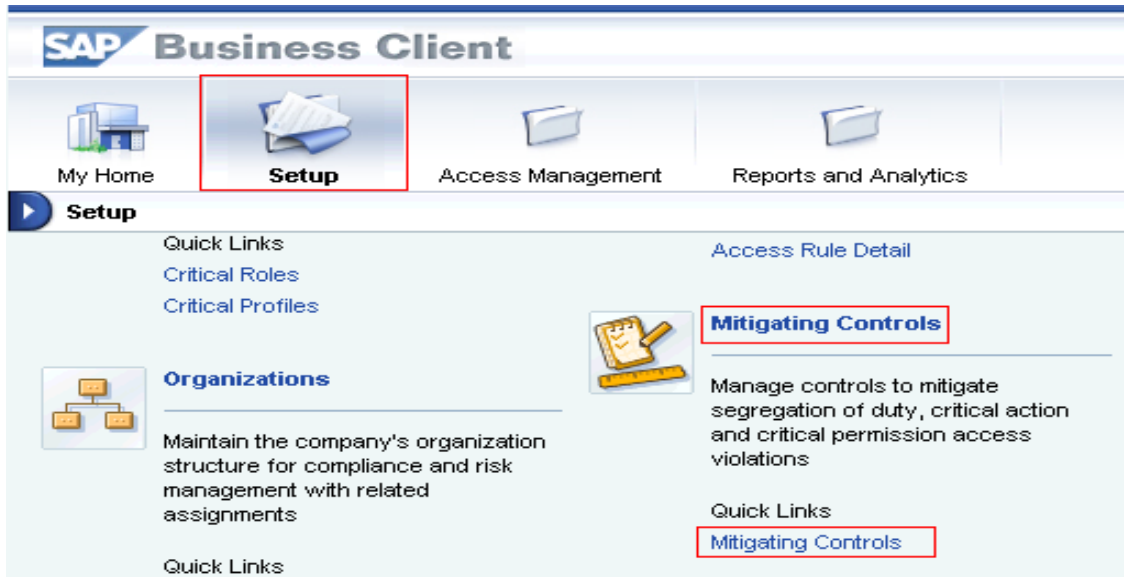


Step 2: Creating new Functions.

Choose setup work center->*Access Rule Maintenances* → *Functions*

Creating mitigation controls.

Here, we can manage controls to mitigate segregation of duty, critical action and critical permission access violations.



The screenshot shows a web browser window displaying the 'Mitigating Controls' page. The page has a header with the URL 'http://india.server.com:8000 - AC Mitigating Control - Microsoft Internet Explorer'. Below the header, there is a section titled 'Active Queries' with a sub-section 'Mitigating Controls All (3)'. The main content area is titled 'Mitigating Controls - All' and includes a 'Show Quick Criteria Maintenance' button. Below this, there is a table with the following columns: Control ID, Title, Description, Organization, Last Updated, and Updated By. The table contains three rows of data.

Control ID	Title	Description	Organization	Last Updated	Updated By
M001	dfydr	dfydr	IBM	25.04.2012	PURNASRI
MIT10	MIT10	MIT10	GRC	04.08.2012	PURNASRI
MM02	vcsdavc	vcsdavc	IBM	20.07.2012	PURNASRI

All the Mitigation Controls, corresponding Access Risks, Owners, Reports and Attachments have to create Manually, by clicking "Create" as shown below.

http://india.server.com:8000 - Object Information - Microsoft Internet Explorer

Control

Save Cancel

General Access Risks Owners Reports Attachments and Links

Mitigating Control ID: *

Name: *

Description:

Organization: *

Process:

Subprocess:

Notes

B Z H1 H2 H3

Need filled the required details as per below screen

General Access Risks Owners Reports Attachments and Links

Mitigating Control ID: * MM03

Name: * MT03

Description: hghgi

Organization: * IBM

Process: Basis

Subprocess: client administration

Notes

General Access Risks Owners Reports Attachments and Links

Access Risks

Add Row Remove Row

Risk ID	Rule ID	Description	Level	Updated By	Updated On
RSK001		basisrisk	Medium	CHERRYBABU	13.08.2012

Save Cancel						
General Access Risks Owners Reports Attachments and Links						
AC Owners Add Row Remove Row						
	Name	Assignment Type	Type	User ID	Updated By	Updated On
	CHERRYBABU	Approver	Owner	CHERRYBABU	CHERRYBABU	13.08.2012
	CHERRYBABU	Monitor	Owner	CHERRYBABU	CHERRYBABU	13.08.2012

Running the risk analysis

Choose Access management work center->Access Risk Analysis→user level.

The screenshot shows the SAP Business Client interface. The top navigation bar includes 'My Home', 'Setup', 'Access Management', and 'Reports and Analytics'. The 'Access Management' section is active, displaying a sidebar with 'Access Risk Analysis', 'Access Request Administration', and 'Role Management'. The main content area shows 'Access Risk Analysis' with a description and quick links. Other sections like 'Mitigated Access', 'Access Request Creation', and 'Compliance Certification Reviews' are also visible.

The screenshot shows the 'Risk Analysis : User Level' configuration window. It includes sections for 'Analysis Criteria' (System, User, User Group, Custom Group, Rule Set, User Type), 'Report Options' (Format, Type, Action Level, Permission Level, Critical Action, Critical Role/Profile, Critical Permission), and 'Additional Criteria' (Include Mitigated Risks, Show All Objects, Offline Data). There are buttons for 'Run in Foreground', 'Run in Background', and 'Save'. The 'Saved Variants' section shows a list of variants with a 'Delete' button.

The result not only shows the list of Risk IDs, Function IDs, Business Processes but refines the result and shows the Actions and Permission Objects (at Field values) as well.

Display As: <table><tr><td>Table</td><td>Print Version</td></tr></table>													Table	Print Version
Table	Print Version													
Risk ID	Access Risk Description	Access Risk Type	Access Risk Level	Business Process Description	Functions	Function Description	Action	Access Rule ID	Permission Object	Field	Status	Value From	Value To	
	Basis Development & Configuration	Segregation of Duties	High	Basis	BS02	BS02 - Basis Development	SE13	00GU	S_DEVELOP	P_GROUP	InActive	*		
	Basis Development & Configuration	Segregation of Duties	High	Basis	BS06	BS06 - Configuration	SM49	00GU	S_TCODE	TCD	Active	SM49		
	Basis Development & Configuration	Segregation of Duties	High	Basis	BS06	BS06 - Configuration	SM49	00GU	S_LOG_COM	COMMAND	InActive			
	Basis Development & Configuration	Segregation of Duties	High	Basis	BS06	BS06 - Configuration	SM49	00GU	S_LOG_COM	HOST	InActive			
	Basis Development & Configuration	Segregation of Duties	High	Basis	BS06	BS06 - Configuration	SM49	00GU	S_LOG_COM	OPSYSTEM	InActive			
	Basis Development & Configuration	Segregation of Duties	High	Basis	BS06	BS06 - Configuration	SM49	00GU	S_RZL_ADM	ACTVT	Active	01		
	Basis Development & Configuration	Segregation of Duties	High	Basis	BS06	BS06 - Configuration	SM49	00GU	S_RZL_ADM	ACTVT	InActive	03		

Once you find the risks if you want to mitigate the user by click on the mitigate risk button.

Mitigation											
Assign Mitigation Controls											
Add Remove Status Validity Period Create Control View Details											
User ID	Org Rule ID	Access Risk ID	Description	Rule ID	System	Control ID	Monitor	Valid From	Valid To	Status	
ARA1		RSK001	basisrisk	*	*	MM03	CHERRYBABU	13.08.2012	13.08.2013	Active	▼

Inference: This is the end of ARA configuration in GRC 10.0

